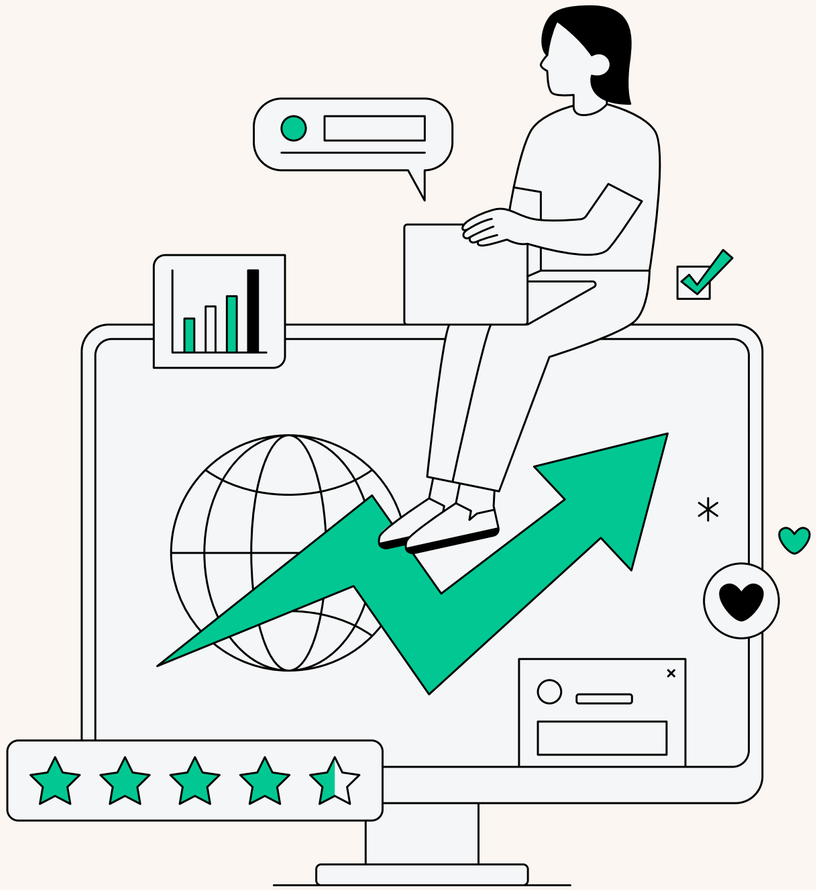
Kevin Baumgartner - Ali Rammal - Fabian Perez-
Abdullah Ghosheh- Nicola Terlikas - Alex Svenburg

Analyses

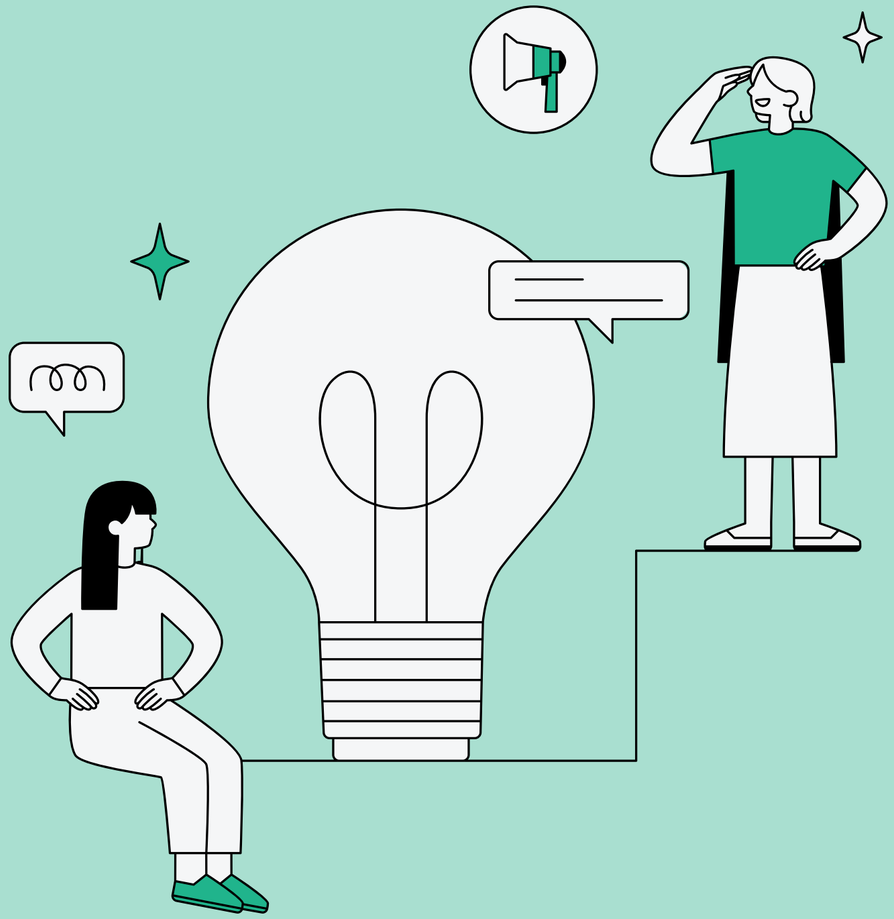
Stream Analytics

GROUP # 1



Introduction to Milestone 2

- Context: Brief introduction to Spotify Wrapped and its significance in user engagement and marketing.
- Objective: To dissect user data through three focused analyses to optimize Spotify's services and user targeting.
- Data Considerations: Highlight the importance of real-time data processing and how it feeds into timely and relevant user insights.



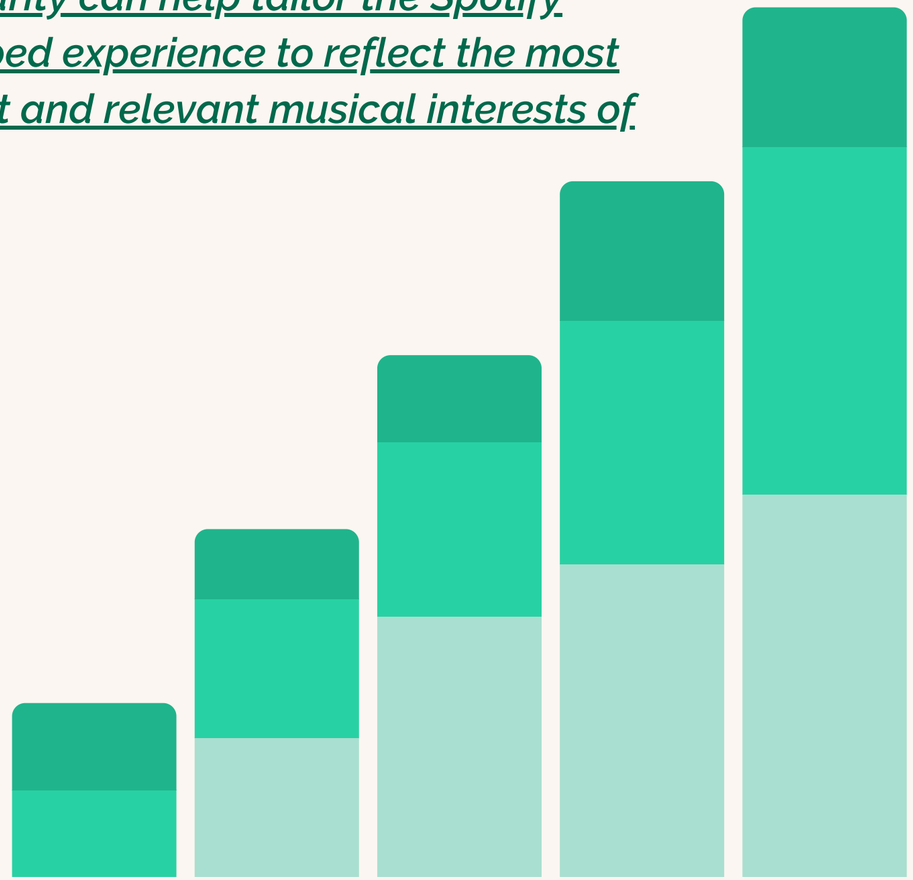
Analyses

Tracking Genre Popularity Over Time

Real-Time Genre Popularity Analysis

- We aimed to track the popularity of music genres over time using real-time data streaming.
- The output as a result streams the results into memory for real-time querying and visualization.
- The use of streaming data and watermarks handles late data effectively, ensuring that the analysis remains up-to-date and accurate.

Knowing real-time trends in genre popularity can help tailor the Spotify Wrapped experience to reflect the most current and relevant musical interests of users.



Genre Popularity by Location

Geographic Insights into Genre Popularity

- Here we worked to determine the popularity of different music genres across various locations using real-time data streams.
- The given output streams the aggregated data into memory for live querying and potential real-time visualization.
- **Windowing and Slide Strategy:** Selected durations provide a balance between timely updates and system resource management.

Relevance to Spotify Wrapped:

- *Localized Content: Understanding regional trends allows Spotify to customize Wrapped features for users based on their location, enhancing personal relevance and engagement.*
- *Strategic Decision-Making: Offers actionable insights for Spotify's marketing teams to focus on areas with high engagement or potential growth.*

Song Popularity by Artist and Location

Musical Trends Across Regions

- Our aim here was to analyze the popularity of individual songs by artists in different locations using a real-time data stream.
- **Listening Duration Calculation:** Determines the exact length of time each song is played.
- **Streaming Data Windowing:** Adopts a 30-minute window with a 15-minute slide to continuously process and update the data.
- **Aggregation:** Data is grouped by song, artist, and location within each time window, with total listening durations summed to measure song popularity.
- Data is then streamed into memory, allowing for real-time querying and analysis.



Relevance to Spotify Wrapped:

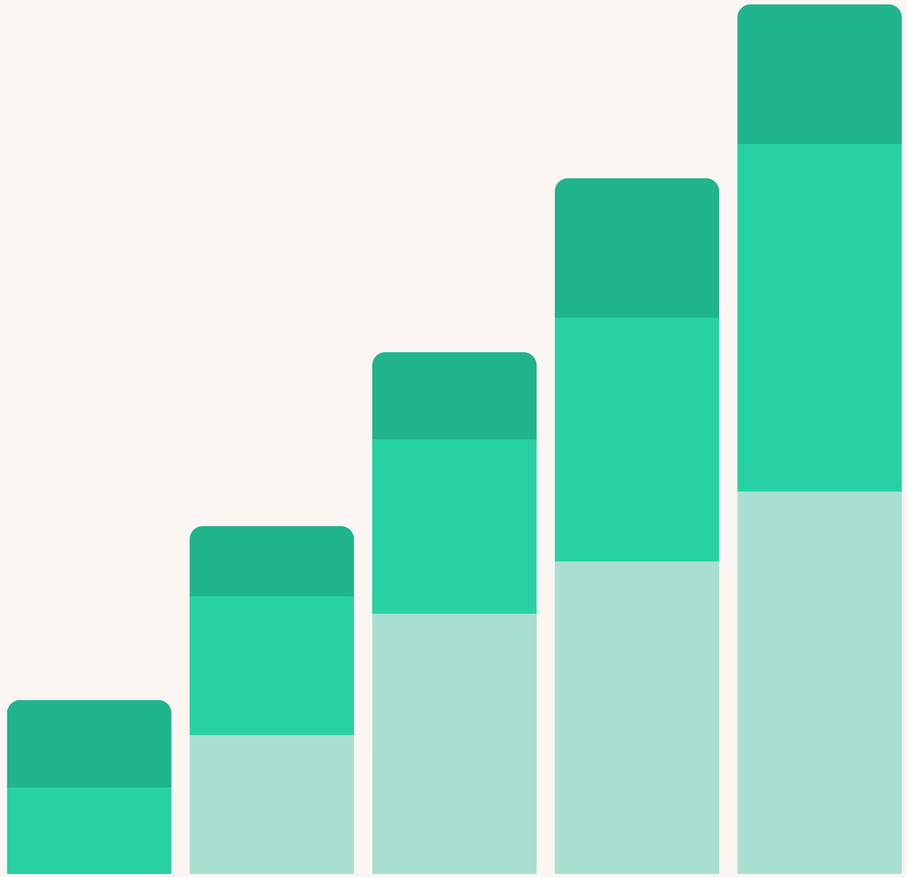
- Personalized User Experience: Offers Spotify detailed data on which songs are favorites in specific areas, potentially influencing user-specific Spotify Wrapped stories.
- Marketing and Promotion: Enables Spotify to design region-specific marketing campaigns based on popular songs and artists.

Considered Analysis

Demographics Query

Demographic Distribution Analysis

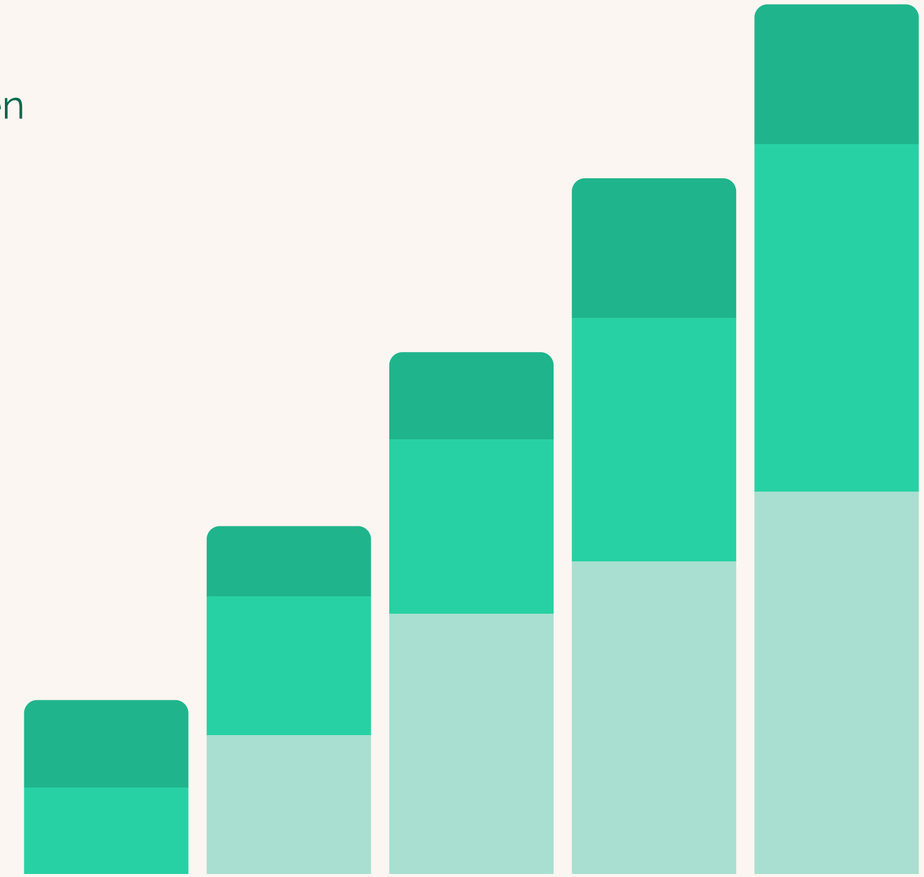
- The idea behind this analysis was to look over the distribution of Spotify users by age and gender.Aimed to provide a basic demographic breakdown, which could help understand user segments more clearly.
- **Reasons for Non-Selection:**
- While demographic data is useful, the query did not offer us the complex insights or real-time relevance required for Spotify Wrapped,
- This analysis could be reused if additional elements and dimensions were present



Listening Habits Query

Time-Based Listening Habits Analysis

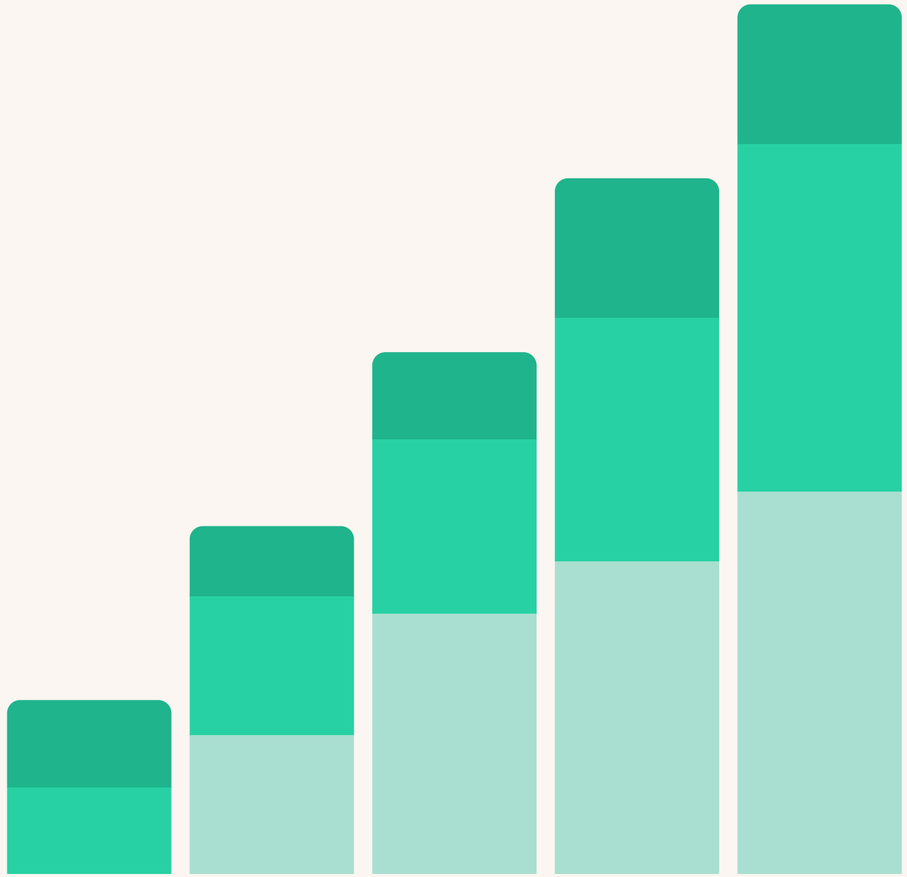
- We decided to track user engagement with music across different times of the day. We intended to identify peak listening times and how listening habits vary throughout the day, categorized into different music time slots with this analysis
- This analysis focuses more on when people listen rather than what they listen to, which may be less relevant for enhancing Spotify Wrapped,
- Aspects of temporal listening habits could be inferred from other, more content-focused analyses which could be a further improvement by integrating more data.



Genre Popularity by Demographics

Demographic Insights into Genre Preferences

We are examining how music genre preferences vary across different age groups and genders.
This option was **not** selected because this analysis might overlap with other demographic-focused queries, possibly offering redundant insights
A way to improve this would be Integration with Listening Duration this could be enhanced by considering not just the count of interactions.



Genre Popularity by Location

Regional Music Preferences Purpose of the Analysis

We aim to determine how music genre preferences vary across different geographical locations.

Reasons for Non-Selection would be that it might overlap with similar analyses focusing on genre and location,

Yet for future consideration we can integrate more detailed user behavior

Average Listening Duration by Genre

Engagement Level Analysis Across Genres

The aim was to calculate the average listening duration for different music genres over time.

In terms of Real-Time Processing, this approach benefits from streaming data, maintaining freshness and relevance of the data.

User Interaction Counts

This analysis was designed to track when people listen to music rather than the specifics of what they listen to.The data would be segmented into different time slots (morning, afternoon, evening, night) to analyze changes in listening behavior. Uses real-time data streaming to ensure the analysis is current and reflects the latest user behavior trends.

Presented by Group # 1

Thank you very much!

